

Large Loads Frequently Asked Questions

April 2026

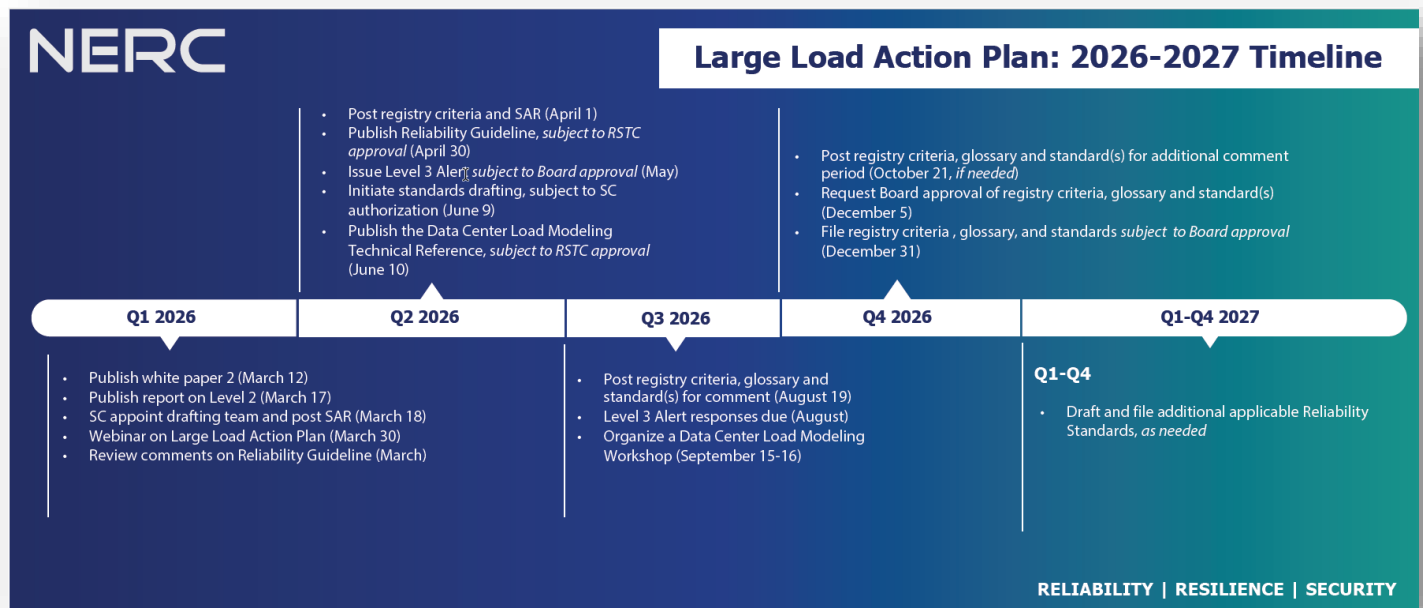
Addressing Large Loads on the Grid

An increasing number of large commercial and industrial loads are rapidly connecting to the bulk power system (BPS). Computational loads, such as data centers (including cryptocurrency and artificial intelligence), hydrogen fuel plants, and others, present unique challenges in forecasting and planning for increased demand.

To begin understanding large loads and identifying effective pathways for their integration, NERC established the Large Loads Task Force in August 2024, now the [Large Loads Working Group \(LLWG\)](#), and developed a [work plan](#); the Member Representatives Committee provided [written input](#) and hosted a [technical panel session](#) at the February 2025 Board meeting; and NERC's Board issued a resolution in February 2025 directing NERC staff to develop an action plan, which will provide additional structure to NERC's efforts related to large loads integration. Through the extensive analysis, report development, and engagement conducted to date, NERC has determined that it must act now to mitigate the risks associated with large computational loads.

What NERC is Doing to Mitigate Risks Associated with Large Loads

NERC's Large Loads Action Plan



Advancing a three-pronged, concurrent approach:

- **Technical Justification.** An Essential Action Level 3 alert will be issued to recommend near-term mitigation actions that current registered entities can take to address critical reliability risks, expected May 2026.

RELIABILITY | RESILIENCE | SECURITY

POWERING TODAY. PROTECTING TOMORROW.

Additionally, the LLWG plans to publish its Reliability Guideline for large load interconnections during the same period, *subject to RSTC approval*.

- **Registration Criteria for Computational Loads.** NERC will define which entities, based on specific physical and electrical criteria, would be required to register with NERC and comply with its Reliability Standards.
- **Reliability Standards.** NERC will revise and adopt new Reliability Standards to define computational loads and establish measurable requirements for newly registered entities, with an initial Reliability Standard completed, expected by the end of 2026.

Actions by NERC to Address Large Loads on the Grid

NEW: Standards Development

The Standards Committee launched [Project 2026-02](#) on March 16, 2026, to develop Reliability Standards addressing computational loads. The committee appointed a drafting team and accepted the Standard Authorization Request (SAR) for posting. The SAR was posted for 30-day formal comment period on April 1, with comments due by April 30.

Additionally, subject matter experts with relevant experience can volunteer through April 30, 2026, for the Computational Load Short-Term Advisory Group. The Advisory Group will support the initial standard project by contributing at least three hours per week to support the drafting team. In addition, observers interested in this project may join public drafting team meetings and actively participate in the process. Additional information is on the [Reliability Standards Under Development](#) page.

NEW: Registration

NERC proposes to establish a new registered entity, subject to Reliability Standards, to address large computational loads. Accordingly, NERC plans to revise its Rules of Procedure, Appendix 5B Statement of Compliance Registry Criteria to account for Computational Load Entities. The draft was posted for a 45-day comment on April 1, 2026 on the [Proposed Changes to Rules of Procedure](#) web page.

Large Loads Working Group

The Large Loads Working Group (LLWG) focuses on understanding the reliability impacts associated with emerging large loads. This includes identifying gaps in existing processes and recommending enhancements to planning and operations. The group will [meet](#) in 2026 on April 2, July 2, and October 1. To join the LLWG, please contact [Evan Mickelson](#).

Load Modeling Working Group

The [Load Modeling Working Group](#) (LMWG) actively engages stakeholders through technical presentations and discussions focused on modeling large loads. Additionally, the group is developing a technical reference document to inform industry on how to model these emerging loads for dynamic studies. That draft will be shared with the RSTC for comment on April 30, 2026. To join the working group, contact [Hasala Dharmawardena](#).

Key Reports and Findings

NERC's LLWG has several ongoing projects including a published white paper on [Characteristics and Risks of Emerging Large Loads](#) which finds that peak demand is just one of several factors that can impact BPS reliability, and further action is needed to address these risks.

NEW: A second white paper examined [Gaps in Existing Practices, Requirements, and Reliability Standards for Emerging Loads](#) and found that the existing NERC Reliability Standards, as well as industry processes and requirements, are inadequate for the reliable integration of emerging large loads onto the BPS. The white paper contains recommendations

for NERC, the LLWG, registered entities, regulators, policymakers, as well as others. These recommendations include updates to registration criteria, Reliability Standards, processes, and requirements.

Reliability Guideline

The draft [Reliability Guideline: Risk Mitigation for Emerging Large Loads](#) was posted on November 3, 2025, for a 45-day industry comment period. NERC received more than 800 comments that will help shape the recommendations for risk mitigation. The focus will be on improving modeling practices, analyses, coordination, data collection efforts, real-time monitoring, and event analysis. NERC is appreciative of the level of stakeholder engagement. The final Reliability Guideline is expected to be published in the second quarter of 2026.

FERC Advance Notice of Proposed Rulemaking

NERC submitted [comments](#) to the Federal Energy Regulatory Commission's Advance Notice of Proposed Rulemaking (ANOPR) addressing the reliable interconnection of large loads. NERC previously filed [comments](#) in the Commission's co-location proceeding in Docket No. AD24-11. In the ANOPR comments, NERC cites its support for addressing the rapid growth of these large loads and appreciation for recognition of NERC's role and its request that future actions reference NERC's full authority under Section 215 of the Federal Power Act to include users, owners, and operators of the BPS. Additionally, the comments underscore that NERC is actively collaborating with stakeholders to assess risks and mitigation options and to provide details on potential future efforts.

In March, NERC submitted a supplemental [letter](#) to FERC detailing an accelerated action plan, including revising our registry criteria and Reliability Standards by the end of 2026. A Level 3 Alert will also be issued this May to kickstart essential risk mitigation actions.

Industry Alert

NERC issued a [Level 2 Industry Recommendation Alert](#) with a January 28, 2026 response deadline. The alert, which was sent on September 9, urges industry participants to take specific actions to mitigate risks to BPS reliability that may arise from the integration of large loads. Specifically, the alert focuses on evaluating dynamic modeling and simulation practices related to large loads and their interconnection requirements. NERC is assessing the study processes, commissioning procedures, and dynamic modeling for large loads to identify additional actions needed to address any observed deficiencies.

NEW: NERC has posted a [report](#) summarizing the key findings from the Level 2 Recommendation to Industry: *Large Load Interconnection, Study, Commissioning, and Operations* with specific actions to mitigate risks to the BPS.

Technical Conference

NERC hosted a technical [conference](#) on February 24–25, 2026, to discuss power system risks and associated mitigation pathways for emerging large loads. The conference focused on data centers and the characteristics of computational loads. This conference also discussed planning and operations, interconnection processes, facility performance requirements, and other topics related to the integration of large computational loads.

How do I stay informed?

Large Loads Resources

NERC has established a dedicated [web page](#) under the Initiatives tab on www.nerc.com to provide information on the Large Loads Action Plan. The page features resources from the [LLWG](#) and [LMWG](#) as well as [Quarterly Updates](#), [meeting presentations](#), [recordings](#), and more information.

About Large Loads

What characteristics are typical for large loads?

Historically, large loads referred to industrial facilities with high electrical demand. They had longer interconnection¹ timelines that allowed for more study time under traditional planning processes. Currently, emerging large loads include cryptocurrency mining, data centers (conventional and artificial intelligence), which may be referred to as computational loads. Emerging large loads also include oil field loads and hydrogen production facilities. Many have a shorter timeline to interconnect (months vs. years) to the grid. In addition to these rapid timelines, some emerging large loads introduce new challenges to grid operators like rapid demand fluctuations and increased voltage sensitivity.

What challenges exist with integrating emerging large loads onto the grid?

Integrating emerging large loads onto the grid poses several challenges, including accurately forecasting future demand, ensuring that transmission and generation capacity keeps pace with this demand, and managing rapid fluctuations in consumption during all conditions – both fault and normal – which can destabilize the grid.

What Reliability Challenges do Large Loads Pose to the Bulk Power System?

NERC has observed several reliability challenges resulting from large loads around timing, data collection and forecasting, and modeling.

Timing

Some large loads seek to connect within one-to-two years, and the traditional planning processes are not equipped to ensure the grid can reliably serve this new demand. Additionally, the transmission upgrades required to handle the full new requested loads cannot be completed quickly enough. Since some loads want to connect to the BPS as soon as possible, they submit speculative interconnection requests to find the interconnection point with the fastest timeline, significantly increasing the number of interconnection studies utilities must perform. Due to the sheer number of interconnection requests, limited time is available to study the implications of these new loads.

Data Collection and Forecasting

To better quantify the risk, the electrical characteristics of large load facilities, such as ramping capabilities, power electronic settings, internal protection schemes, and coordination across multiple facilities, must be better understood. A better understanding of standardizing large-load modeling for long-term forecasting is needed.

Modeling

Current dynamic load models have challenges accurately representing the behavior of emerging large loads such as data centers. Considering the immense growth of these loads, modeling their dynamic behavior has become more critical than ever. NERC's Load Modeling Working Group (LMWG) is working on modeling methodologies that can accurately simulate these emerging loads' electrical behavior. These models can be used in studies to assess risks posed by large loads.

Transmission Planning

Currently, transmission expansion projects can take a decade from planning to energization. There is a growing urgency for processes to address the shorter-term needs associated with this rapid load growth. Additionally, rapid integration of large loads may pose short and long-term resource adequacy concerns, as generation can take years to be operational, longer than some large load interconnection timelines of months to years. A proactive, holistic approach is necessary for integrating large loads.

Voltage Ride-Through

¹ Interconnection refers to the process of integrating new generators, loads, or other equipment into the grid.

Currently, there are no specific voltage ride-through Reliability Standards for large loads. Large loads, specifically data centers, frequently house a multitude of sensitive electronics that require ideal electrical conditions. If poor electrical conditions exist (like low voltage), large loads can disconnect from the grid to protect their equipment from damage. This behavior gives rise to grid stability challenges.

Co-Locating Large Loads

Large loads co-located at generating facilities are referred to as co-located large loads. The Federal Energy Regulatory Commission, NERC, and industry began examining co-located loads in 2024, with additional proceedings in 2025 and 2026. Co-located loads present unique considerations as discussed in NERC's filing in that Commission proceeding.

NERC's findings and filings have highlighted the importance of planning and coordination to reduce risks, while noting that overall reliability impacts will depend on how large loads are implemented, their associated technologies, and any applicable regulatory framework. NERC has emphasized that maintaining diversity across electric generation and consumption will help ensure system reliability and resilience. NERC also noted the complexity of load balancing and forecasting associated with the changing resource mix. The complexity of this load and generation balance will continue to increase as more data centers connect to the grid.

Continued assessments reflect other potential considerations. For example, co-location of a generation unit at a load facility (or vice versa) can result in close dynamic interactions between these loads, generation and parts of the connecting utility, even in scenarios where the utility connection is only the provider of last resort. Without planning, co-location could also impact registration status of existing registered entities to expand the functions applicable. These dynamic interactions could affect the reliability of the co-located facilities as well as the other users of the interconnected BPS.

Furthermore, generation capacity and the [Essential Reliability Services](#) that are associated with it are critical to maintaining BPS reliability. When generation capacity is removed or reduced from one location, the Essential Reliability Services it provided need to be supplemented elsewhere. Regardless of co-location or direct connection, sufficient Essential Reliability Services will need to be provided on the transforming grid to maintain BPS reliability. All utility-connected load facilities rely upon the Essential Reliability Services provided by the interconnected bulk power system.